Ledbetter Data Pump (LDP)

User Guide: Writing Pipeline YAML (Phase 1)

For humans and coding agents

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Version 1.0

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Audience: Humans and coding agents

This guide explains how to write a valid `pipeline.yaml` file for Phase 1 of the Ledbetter Data Pump.

It is deliberately precise so both humans and LLM agents can produce correct files.

1. File Structure Overview

Every pipeline file has this shape:

```
version: "1.0"
pipeline:
  name: string
  settings: { ... }          # optional
  resources:
    connections: [ ... ]
  dataflow:
    sources: [ ... ]
    transformations: [ ... ] # optional
    targets: [ ... ]
```

- `version` must be exactly `"1.0"`.
- Exactly one source is supported in Phase 1.
- All `id` values must be unique.
- Every `input` must reference an existing `id`.

2. Connections

```
resources:
  connections:
    - name: "pg"
      type: "postgres"
      url: "env://PG_URL"          # preferred
      # or
      # url: "postgres://user:pass@host:5432/dbname"
```

Rules

• `type` must be `"postgres"` in Phase 1.

• Prefer `env://VARIABLE_NAME`. The variable must be set in the environment.

• Plain URLs are allowed but passwords will be redacted in logs.

• Connection names are referenced by sources.

3. Sources

```
sources:
  - id: "src"
    connection: "pg"
    query: |
      SELECT id, name, status, amount, created_at
      FROM customers
      WHERE created_at >= '2026-01-01'
    partitioning:
      type: "range"          # or "none"
      column: "id"           # required for range
      chunks: 8              # required for range
```

Partitioning

• `type: range` — engine runs `MIN`/`MAX` on the column and splits into `chunks` ranges.

• `type: none` — single partition, entire query result.

• Partition column must be an integer type (BIGINT/INTEGER).

• The connector automatically adds `WHERE column >= $1 AND column < $2`.

4. Transformations (Optional)

Phase 1 supports only `filter` and `derive`.

Filter

```
- id: "active_only"
  input: "src"
  type: "filter"
  condition: "status = 'ACTIVE' AND amount > 0 AND created_at IS NOT NULL"
```

Derive

```
- id: "with_tax"
  input: "active_only"
  type: "derive"
  fields:
    - name: "taxed_amount"
      expression: "amount * 1.0825"
    - name: "processed_at"
      expression: "now()"
```

Phase 1 Expression Rules

• Columns, numeric/string/boolean/null literals

• Comparisons: `= != > >= < <=`

• Logical: `AND OR NOT` (use parentheses)

• Arithmetic: `+ - * /`

• `IS NULL` / `IS NOT NULL`

• Functions: `coalesce(a, b)`, `now()`

• Strings use single quotes: `'ACTIVE'`

• Nulls follow SQL three-valued logic

Not supported yet: `IN`, string functions, aggregates, subqueries.

5. Targets

Parquet

```
- id: "out"
  input: "with_tax"
  type: "parquet"
  path: "./output/customers/"
  compression: "zstd"       # zstd | snappy | none
```

CSV

```
- id: "out_csv"
  input: "with_tax"
  type: "csv"
  path: "./output/customers.csv"
```

- `path` for Parquet is a directory (one or more files will be written).
- `path` for CSV is a file.
- Files are written via temporary names + atomic rename.

6. Settings

```
settings:
  max_threads: 8                    # default 8
  state_file: ".ldp/my_job.json"    # default ".ldp/<pipeline_name>.json"
  max_batch_rows: 65536             # default 65536
  max_batch_bytes: 134217728        # default ~128 MB
```

All settings are optional.

7. Complete Working Example

```
version: "1.0"
pipeline:
  name: "customer_export"
  settings:
    max_threads: 4
    state_file: ".ldp/customer_export.json"
  resources:
    connections:
      - name: "pg"
        type: "postgres"
        url: "env://PG_URL"
  dataflow:
    sources:
      - id: "src"
        connection: "pg"
        query: "SELECT id, name, status, amount, created_at FROM customers"
```

```
    partitioning:
      type: "range"
      column: "id"
      chunks: 4
  transformations:
    - id: "active_only"
      input: "src"
      type: "filter"
      condition: "status = 'ACTIVE' AND amount > 0"
    - id: "with_tax"
      input: "active_only"
      type: "derive"
      fields:
        - name: "taxed_amount"
          expression: "amount * 1.0825"
  targets:
    - id: "parquet_out"
      input: "with_tax"
      type: "parquet"
      path: "./output/customers/"
      compression: "zstd"
```

Expected Postgres table

```
CREATE TABLE customers (
    id BIGINT PRIMARY KEY,
    name TEXT,
    status TEXT,
    amount DOUBLE PRECISION,
    created_at TIMESTAMP
);
```

8. Common Validation Errors

9. Phase 1 Limitations (Do Not Use Yet)

• Multiple sources

• Conditional splits
• Snowflake, SQL Server, Oracle
• `IN` lists, string functions, complex expressions
• Incremental / watermark loads
• Atomic swap or merge modes
• Custom null handling or encoding options

10. Quick Checklist for Agents and Humans

• [ ] `version: "1.0"`
• [ ] Unique `id`s
• [ ] Every `input` resolves
• [ ] Postgres connection uses `env://` when possible
• [ ] Partition column is integer
• [ ] Expressions use only allowed operators and functions
• [ ] Parquet `path` is a directory
• [ ] State file path is writable

End of User Guide

A file that follows this guide will pass Phase 1 validation and execute correctly.